Department of Mechanical and Mechatronic Engineering
Stellenbosch University

CNC Router Procedures

Mr K Neaves
May 2021, version 1.0

1. Introduction

This document outlines the procedures for making use of the CNC router for skripsie and postgraduate use. Firstly, some important decisions need to be made when considering the use of the CNC router. Thereafter, the specifications of the available router and drawing and setup processes are discussed. Lastly, the procedure for using the CNC router facilities is described.

2. Suitability for CNC routing

There are many aspects to consider when designing a component, and several of these aspects directly influence the manufacturing method that will be required. Some points to consider before deciding on using the CNC router:

- Can the component be purchased off the shelf?
- Can the component easily be manufactured through another available process?
- How many parts will be manufactured?
- Material choice.
- Can the cost of manufacture be justified by the intended function of the component?
- Can the part easily fit within the machine dimensions, specifically (z-axis).
- What tooling would be necessary to machine the part? Is that tooling available?

3. Router specifications

This section covers some of the basic router specifications as well as considerations for the design and manufacturing of the part.

3.1. Routing process

A CNC router is a computer controlled cutting machine that forms the desired part from stock by the process of material removal. In many ways a CNC router is similar to a CNC milling machine, however far less rigid. A typical CNC router consists of 3 axes and a spindle. The spindle is what holds and rotates the cutting tool necessary for the work. The machine can be controlled manually, however in most circumstances a CAM (computer-aided manufacturing) program will be used to create the necessary toolpaths from a CAD model. The CAM software will produce a G-code file which the CNC controller reads in order to move and control the machine in the programmed contours.

3.2. CNC Router

The facility has one UNICAM CNC router (Figure 1). The router specifications are summarised in Table 1, but may differ slightly from the values stated.

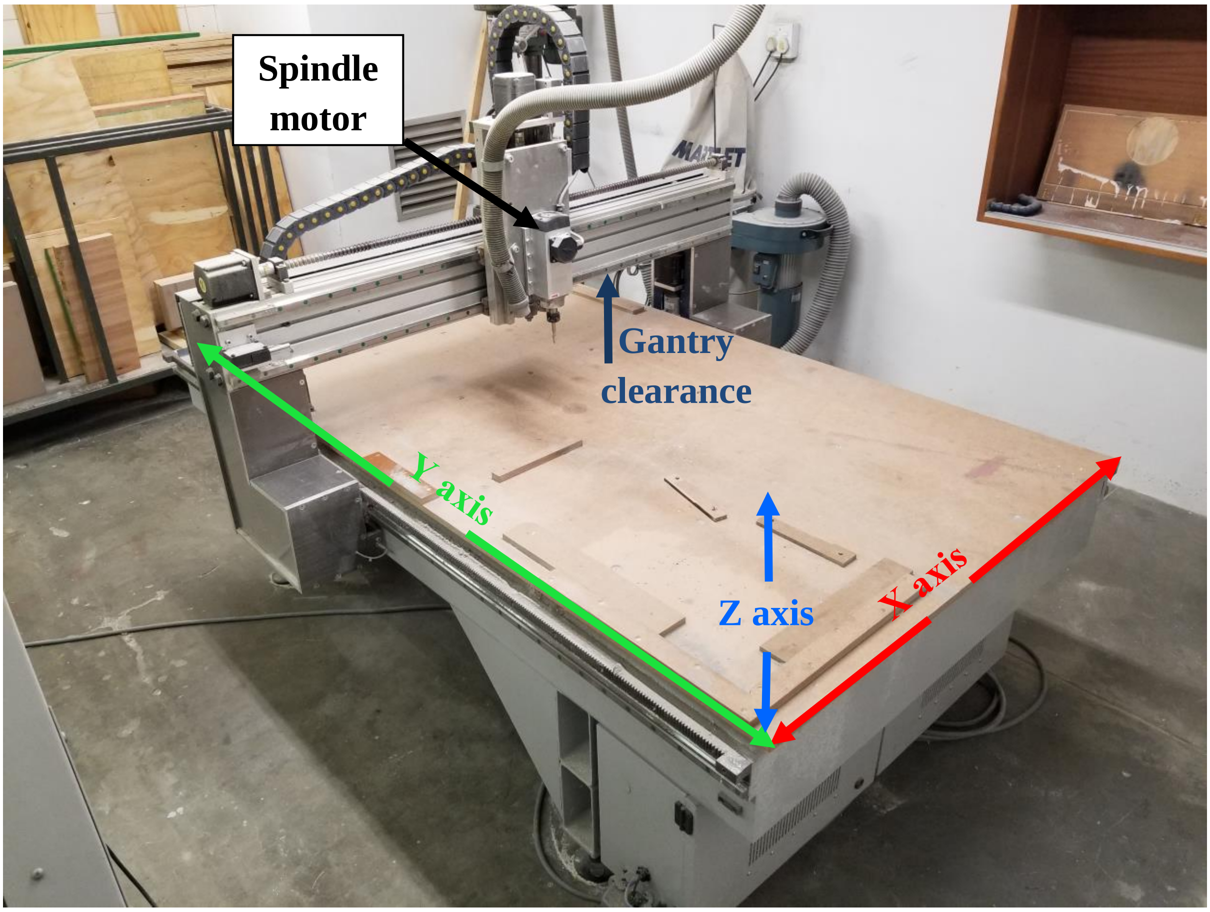

Figure 1: UNICAM CNC router.

Table 1: UNICAM CNC router specifications

Bed size	1220 mm by 1800 mm (x, y)
Gantry clearance	< 190 mm (approximately)
Spindle collet to bed clearance	< 215 mm (approximately)
Max travel speed	10000 mm/min
Spindle speed	8000 to 24000 rpm
CNC machine controller	Mach3
Motion controller	CSMIO/IP-M
Program language	G-code

3.3. Design and manufacturing considerations

Special care should be taken in the design of parts to make them suitable for CNC routing and to ensure clarity and ease of set up for the artisan. The most important design aspects are discussed in this section.

3.3.1. Level of detail

It should be realised that fine detail in a model will often require small tooling and possibly increased machining time. For example, the profile shown in yellow in Figure 2 would be difficult to machine exactly as per the model due to the curve and that the wall is perpendicular to the wheel arch with no fillet in the model. Obtaining the sharp corner as it is in the model will be difficult and would require fine tooling to achieve. Tool access limitations would then most likely become an issue. Should this be machined with a ball endmill there would be at least one tool radius formed in that area (a similar example showing the position of the radius is shown in Figure 3c).

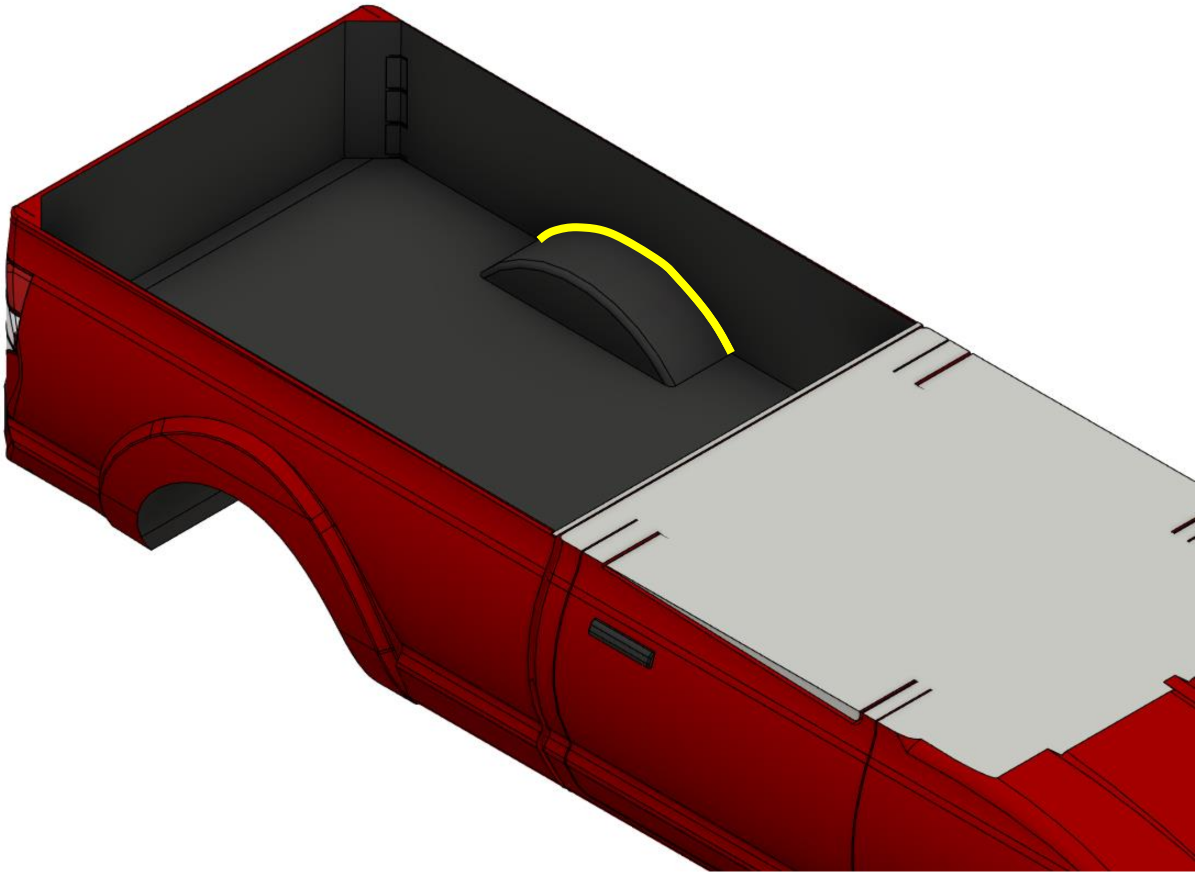

Figure 2: Level of detail example.

Consider the detail necessary for your final part to be functional and choose tooling sizes and number of machining operations accordingly.

3.3.2. Tooling and tool changes

It is key to select a suitable tool type and size for the operation/s that are to be performed. Generally, for softer materials, a tool with a large diameter can be used for any clearing operations and depending on the part geometry it is also possible to perform finishing operations without a tool change. As the machine does not have an automatic tool changer it is preferred to keep tool changes to a minimum and rather only switch tools for completely separate operations. For further advice on tooling availability please consult with Graham Hamerse (ghamerse@sun.ac.za) (room M106).

3.3.3. Tool accessibility and clearance

As the machine only has 3 axes it is not practically possible to machine faces that are recessed or beyond perpendicular without specialised tooling or manually rotating the stock.

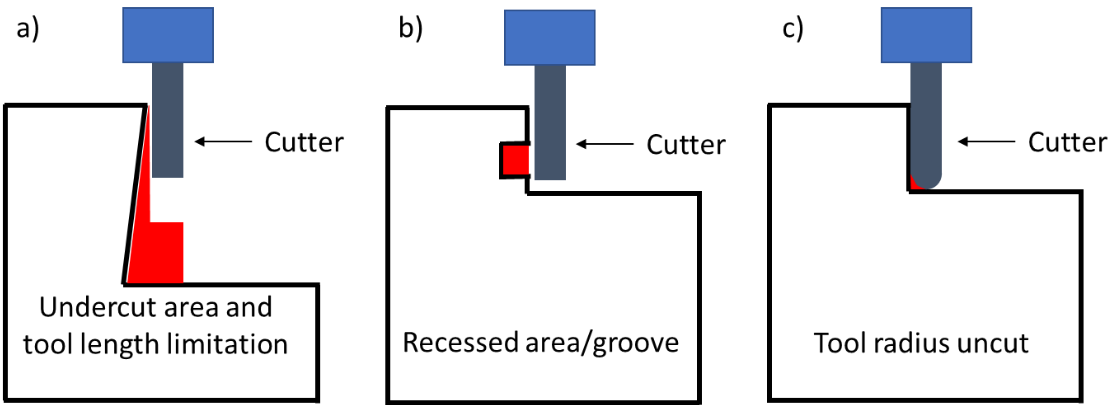

Figure 3: a) Undercut and length limitations. b) Recessed area/groove access. c) Radius from tool uncut.

Another major consideration when preparing a design for machining is the possibility of collisions due to spindle backplate geometry and or tool length (Figure 4 and Figure 3a). These limitations become apparent on parts with steep contours or deep areas that need to be machined. The overall height of the part in combination with the clearance of the tool mounted in the collet must be considered. For example, a part with fine detail in a deep pocket would most likely require specialised tooling and may be beyond the capabilities of the department's machine.

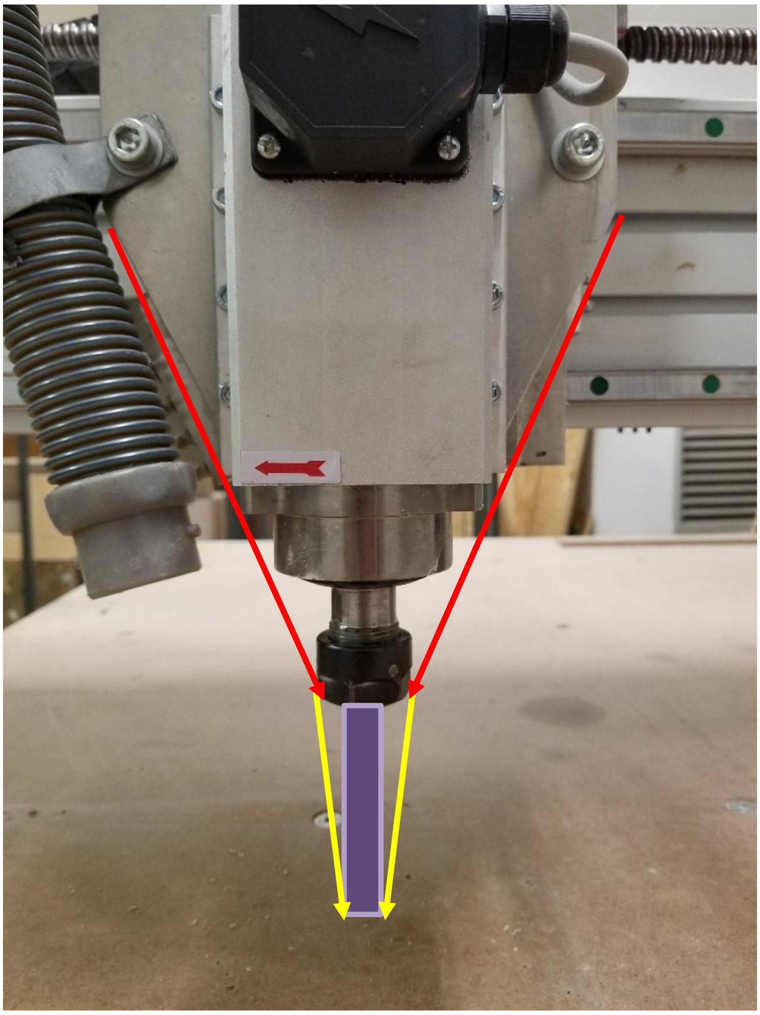

Figure 4: Spindle, spindle backplate (red arrows) and tool length limitations (yellow arrows, tool represented in purple).

3.3.4. Component splitting

It may be necessary to split a part in two or more 'layers' should it be too tall to pass under the gantry and/or under the tool mounted in the collet. An example of this is shown in Figure 5. Note that the layers would need to be aligned and joined in some way after machining has been completed.

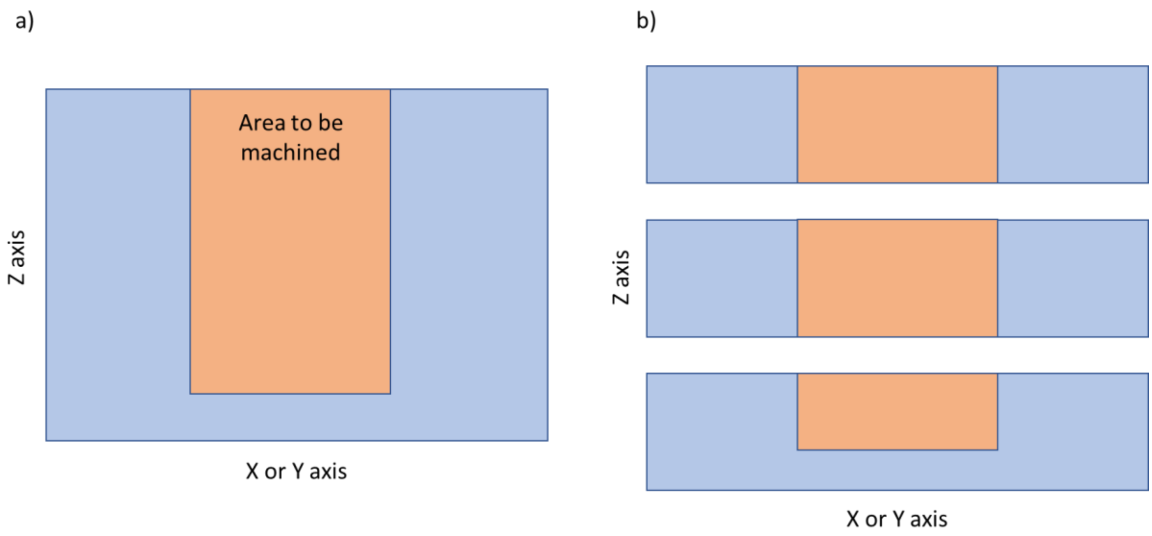

Figure 5: Example of splitting a part into 3 layers such that the tool will not collide with the part and to allow the tool to clear the stock before and during machining. a) Original part. b) Split part.

3.3.5. Work holding

Work holding is often overlooked during the design process, however it is critical in order to obtain a quality end product. Apart from typical working holding clamps (step clamps), several options exist for the CNC router. A frequently used method is that stock can be fastened to the machine bed using wood screws and blocked in as to prevent lateral movement. The screws should either be placed outside of the machining area or counterbored deep enough as to avoid cutting the screws during one of the machining operations.

3.3.6. Material type

The CNC router is capable of machining a variety of softer materials. Woods, tooling board and foam are frequently cut on the machine. Material choice is the designer's decision and should be discussed with Graham Hamerse (ghamerse@sun.ac.za) (room M106) to ensure that it is feasible. Very small and shallow cuts are possible in aluminium, however not recommended.

3.3.7. Support or tabs

Depending on the part geometry, tabs or support may be necessary to keep the part stable (from excessive vibration and deflection during machining and prevent the part from completely breaking loose. Tabs are more frequency required when a part is to be machined from both sides, for example, an aerofoil (example shown in Figure 6). Tabs may also be needed if at any point a majority of the stock has been removed between the part and the work holding fixtures. Tabs are usually removed manually once the CNC operations have been completed.

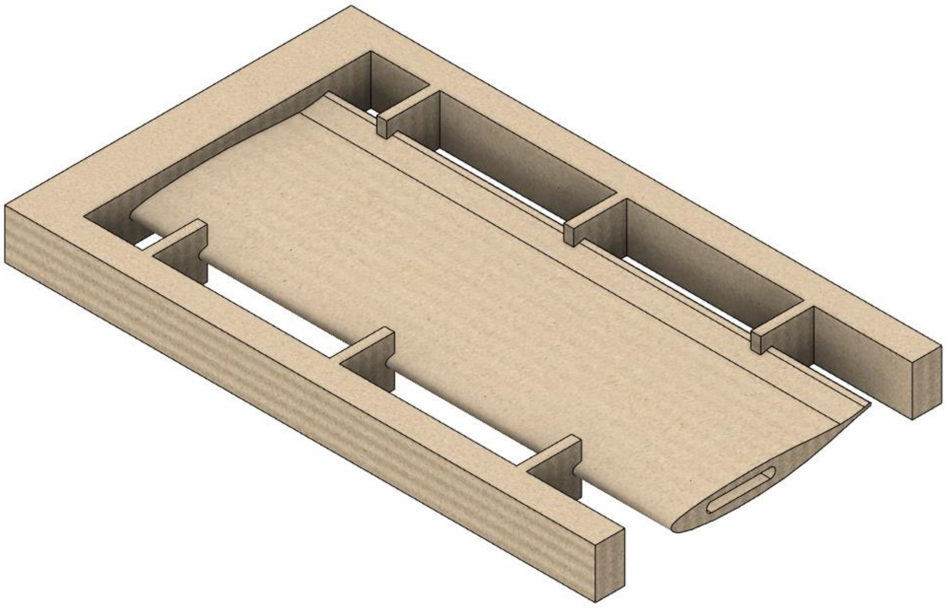

Figure 6: Example use of tabs or supports.

3.3.8. Stock size

Stock is raw material that the part will be manufactured from. Considerations when ordering stock are; allowance for squaring and sizing, work holding and the necessity for tabs or supports. When ordering, attempt to have the stock cut approximately the correct size, allowing for +/- 5 mm (if cut straight) in order to have the stock squared. In almost all cases the workshop will first manually machine and square the stock to a size specified by the designer.

3.3.9. Finishing/post processing

Although a fine finish can be obtained by the correct CAM finishing operation, post processing may still be required due to material limitations. Depending on the material sanding and painting can greatly improve the overall smoothness and finish. Motor car 'body filler' can also be used for covering large voids and re-machined if necessary.

4. Procedure

The procedure that must be followed through the design, preparation and request stages, is outlined in this section.

4.1. Meet with an artisan

It is recommended to meet with Graham Hamerse (ghamerse@sun.ac.za) (room M106) before committing to creating the toolpaths in Fusion 360. Have the following prepared:

- Ensure that supervisor has reviewed the parts
- Reason for using CNC router
- CAD drawings
- Material choice
- At this point you can enquire about cutter options if your CAD models are unlikely to significantly change

4.2. CAM preparation

Parts can be designed in Autodesk Inventor, Autodesk Fusion 360 or your preferred CAD package. The .IPT file can then be opened in Fusion 360. Fusion 360 is used as the CAM software as it is free to students and there is a large amount of information online (YouTube, etc.) on how to set up and use the software.

In Fusion 360 select the "MANUFACTURE" workspace and create a new "Setup" under the milling section. This will bring up the "Setup" dialog where you can specify your work coordinate system (WCS), set your stock size and adjust some initial parameters for the post processor.

Use Figure 1 to aid in orienting the WCS for the manufacturing processes. Simplest is to have the WCS related to the bottom left hand corner of the machine. The WCS origin can either be on top of the stock or on the machine bed, either is acceptable but it is critical that the artisan be informed which method was used and where the origin is (example setup shown in Figure 7). The artisan manually 'zeros' all axes, touching the installed cutter on the relevant surfaces as specified by the designer.

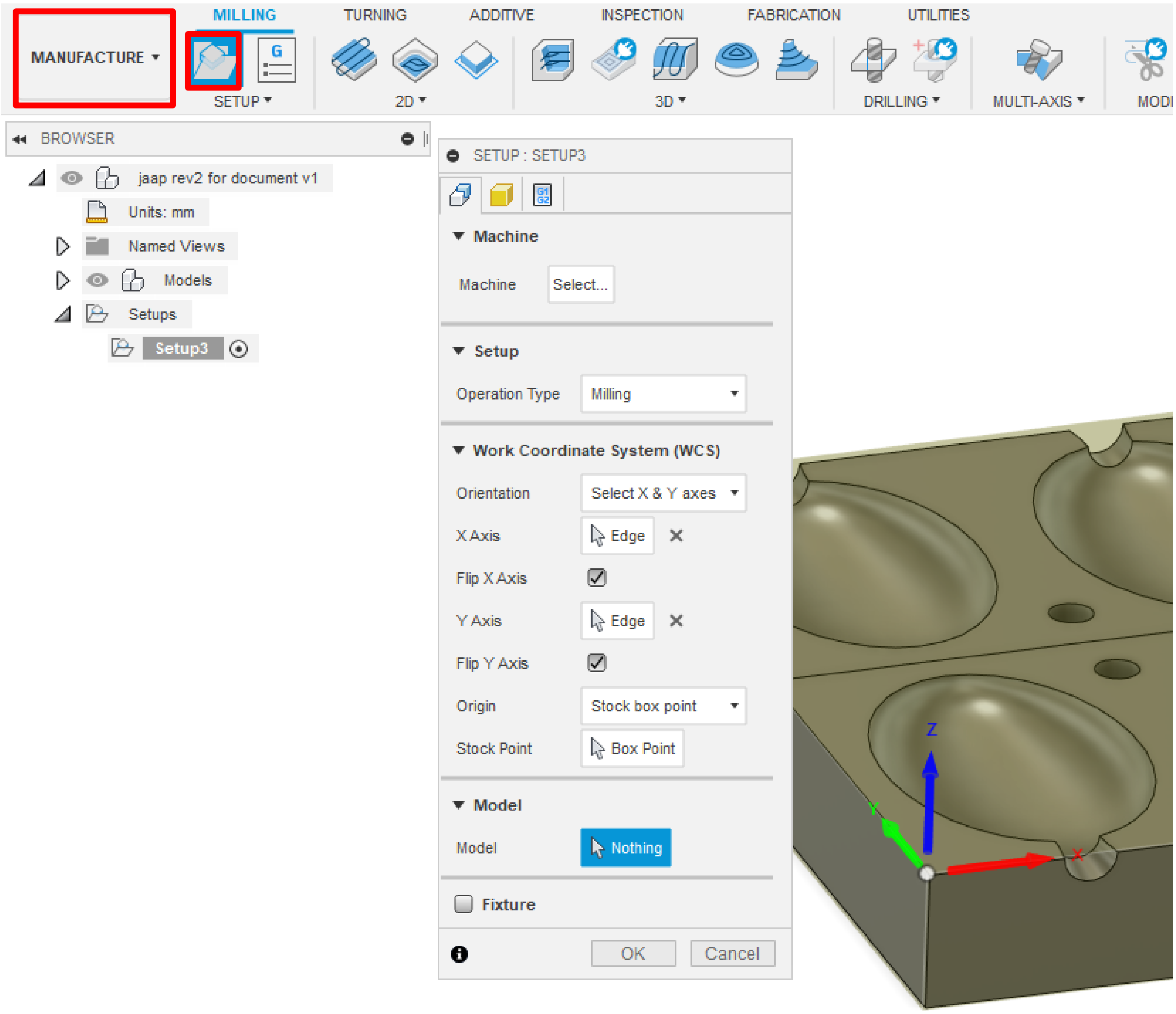

Figure 7: Example of WCS setup.

The stock size can be defined in the same window. An example is shown in Figure 8 where the stock is ‘on size’ with the part, however various options are available should the designer need to add additional stock.

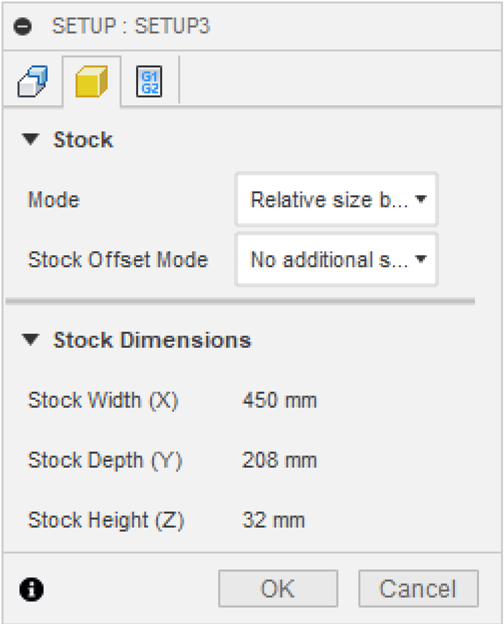

Figure 8: Example of stock setup

Lastly, some initial parameters used for the post processor can be defined. Only the WCS must be changed here and set to a value of “2” (Figure 9).

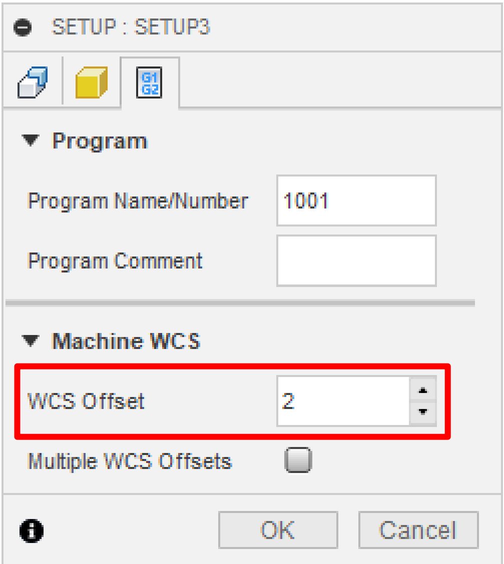

Figure 9: Ensure “WCS Offset” is set to "2".

4.3. Adding custom tools

There is often a need to add a custom tool to the tool library due to differences in the sample and actual cutters available. This is simply done by either copying a similar tool or creating a new tool in the library. After consulting with Graham Hamerse (ghamerse@sun.ac.za) (room M106) of what cutters are available and suitable for the job, it is usually necessary to take measurements of the cutter and enter it into the custom tool you have created. This is necessary to avoid potential collisions or ‘false’ collisions during simulation.

4.4. Machining operations

Machining a part on the CNC router is usually completed in two operations. The first being a roughing or clearing operation and the second being the finishing operation. The purpose of the clearing operation is to remove as much material as possible in the shortest amount of time. This is typically performed using the pocket or adaptive clearing operations found in Fusion 360. There are several important parameters to be set for these operations which should be discussed with Graham Hamerse (ghamerse@sun.ac.za) (room M106). Items to discuss include the maximum roughing stepdown, radial and axial stock to leave (this is excess material that will be removed in the finishing operation), spindle speed and feedrates. Many of these parameters will depend on the selected stock material type.

There are quite a number of options available for the finishing operation and these should be selected based on which operation suits your part geometry the best. On occasion it may be necessary to use multiple finishing operations. This often occurs when a single finishing operation does not yield the desirable finish or is unable to correctly machine all the features of the part.

Many of the finishing operations have a “Stepover” distance parameter. This greatly affects your finished surface roughness and should be chosen relative to the tool diameter you are using. If this is set very small the machining time will become unreasonably long. A stepover of between 0.5 and 1 mm should yield reasonable results on most ball end mill cutters larger than 5 mm diameter.

If possible, it is recommended to experiment generating toolpaths on fewer features as generating the toolpaths can be time consuming on a slower computer. Once the designer is happy with the toolpath type they can then apply it to the whole model or all features.

4.5. Simulation

It is critical that the designer runs the built-in simulation to guarantee that the toolpaths behave as expected and that there are no tool, spindle or machine collisions. If there is an issue in the simulation it is usually highlighted in red on the timeline (provided that the cutter dimensions are specified correctly in the tool library). The simulation will also provide a very rough estimation of the machining time.

4.6. Part checking and scheduling

Once the toolpaths have been refined and checked please share the Fusion 360 project with Kevin Neaves (kneaves@sun.ac.za) via Fusion 360 and also notify Mr. Neaves by email. It may be necessary to schedule a meeting to discuss any problems or intricacies relating to the parts. Should minor or no changes be necessary, schedule the machine with Graham Hamerse (ghamerse@sun.ac.za) (room M106).

4.7. Post processing

This is the last step that should be performed in Fusion 360 and will generate the necessary G-code for the router to use. A post processor simply adapts the G-code to a format that is compatible with the CNC controller that is being used. In this case the post processor that should be selected is “Mach3Mill” (Figure 10).

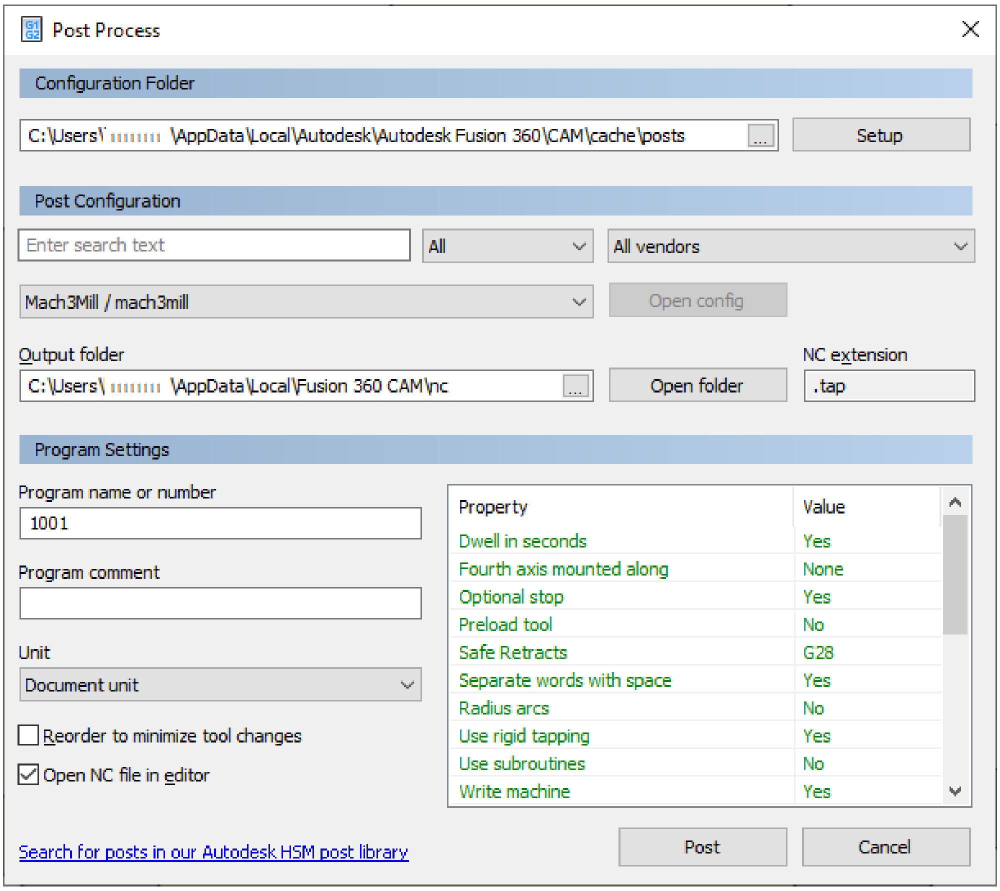

Figure 10: Post process window.

4.8. CNC Machining the part

The final step in the process is to actually begin machining the part. This will happen once the machine has been booked and the stock prepared and fixed to the machine. The G-code should be prepared and copying to a flash disk. The artisan will then load the code, zero the machine axis and begin cutting. Note that you may be required to sit nearby as to stop the machine should something go wrong.